



SDEA PART 2 - Interacting a pilot

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Element 1: [Listen and read back](#)

The second part of the SDEA exam consists of different situations in which the candidate should listen and read back what was informed by the controllers.

Áudio 1: Anac123, climb and maintain 5000 feet, after passing 3000 feet turn left heading 150 and contact departure at 122.40.

Candidate's readback: Anac123, climb and maintain 5000, after passing 3000, turn left heading 150 and contact departure 122.40.

Element 2: [Listen and interact and follow the instructions](#)

The examiner will propose a problem in which pilots should interact appropriately as follows:

Examiner: Now during the climb you realized that your oil temperature is dropping fast, contact the departure control, inform them about your problem and say your intentions.

Candidate: Departure control, Anac123, Pan Pan Pan, my oil temperature is dropping fast, request immediate return to the airport with vectors to ILS.

Áudio 2: Anac123, descend 2000 feet, I will provide vectors to final approach, turn left heading 050, you are cleared ILS Y RWY 02C, confirm you have a fast decrease of your hydraulic pressure?

Candidate: Anac123, descend 2000, turn left heading 050, cleared ILS Y RWY 02C, negative, I am experiencing a fast drop of my oil temperature.



Element 3: Reported speech

Examiner: What did the controller say?

Candidate: First, the controller told me to descend 2000 feet, also informed me he would give me vectors to final approach, right after he instructed me to turn left heading 050, then informed I was cleared ILS Y RWY 02C, and finally asked me to confirm if I had a fast decrease of my hydraulic pressure.

Here the examiners want to check your ability to perform a reported speech. Which actually means they want to assess your ability to say what someone else said, maintaining all information and in accordance with the reported speech rules.

Check here a thorough explanation:

When the reporting verb is in the present tense, then we usually change the tenses in the reported speech, remembering that sometimes the (that) is optional, if you are in two minds of it, just use it.

The technique is based on two parts: the reporting speech and the reported speech. The reporting speech is the introduction that should be created by the candidate, but in general he/she could use some of the following: the controller told me to; the controller asked me to; the controller instructed me to; the controller informed that; the pilot informed that; the pilot reported that...

Reporting Speech

Reporting speech is the introduction of the entire sentence and it varies according to the situation. Sometimes you have more than one option, but one specific reporting speech cannot be used for all situations. Nonetheless the following ones will be enough for you to successfully accomplish the introduction of the reported part of the test.

Examples

Reporting speech what was said by the controllers:

The ATC told me to... The ATC cleared the pilot to... The ATC asked me to...
The ATC informed the pilot that... The ATC asked me to

Reporting what was said by the pilots:

The pilot told the ATC that... The pilot asked the ATC to... The pilot informed
the ATC that... The pilot requested to...

Reported Speech

Reported speech is what was reported by pilots or controllers. The candidate should not change what was said, nevertheless they should perform the shifting time of the verbs used. And the reason is that whenever they do it, the message conveyed will remain the same, here are some examples on how to shift verbs and adverbs.

Examples

Present: understand -> understood

ATC: Anac123 I understand you want to descend

Examiner: What did the controller say?

Candidate: The controller told me that he understood I wanted to descend.

Present continuous: we are -> they were and: I am -> He was

Pilot: Tower, Anac123, we are living FL100.

Examiner: What did the pilot say?

Candidate: The pilot informed the tower that they were living FL100.

Pilot: Approach, Anac123, I am leaving FL100 - The pilot informed the tower
that he was leaving FL100.

Examiner: What did the pilot say?

Candidate: The pilot informed the tower that he was leaving FL100.

Simple past: understood -> had understood

ATC: Anac123, I understood you had an engine failure.

Examiner: What did the controller say?

Candidate: The controller told me (that) he had understood (that) I had
had an engine failure.



Past continuous: I was -> he had been and: saw -> had seen

Pilot: Center, Anac123, when I was passing FL060 and saw a vulture.

Examiner: What did the pilot say?

Candidate: The pilot informed Center that when he had been passing FL100 they had seen a vulture.

Present perfect: haven't seen -> hadn't seen

Tower: Anac123, I haven't seen any aircraft in the circuit.

Examiner: What did the controller say?

Candidate: The controller told me (that) he hadn't seen any aircraft in the circuit.

Past perfect: had taken -> had taken; took off -> had taken off

Tower: Anac123, confirm you said you had taken the runway before the Tucano took off?

Examiner: What did the controller say?

Candidate: The controller asked me to confirm if I had taken the runway before the Tucano had taken off?

Future: will -> would

Approach: Anac123, I'll contact you later.

Examiner: What did the controller say?

Candidate: The controller said (that) he would contact me later.

Modals: would -> would, can -> could, may -> might; shall -> would; must -> must; should -> should.

(would -> would)

Pilot: Approach, Anac123, I would proceed en route, but I am running low on fuel, I have to return.

Examiner: What did the pilot say?

Candidate: The pilot said (that) he would proceed en route, but he was running low on fuel, and also that he had to return.



(can -> could)

Approach: Anac123, I can not vector you, keep holding.

Examiner: What did the controller say?

Candidate: The controller told me (that) he could not vector me, and instructed me to keep holding.

(may -> might)

Approach: Anac123, I may ask you to perform a holding due traffic.

Examiner: What did the controller say?

Candidate: The controller told me that he might ask me to perform a holding due traffic.

(shall -> would)

Pilot: Tower, Anac123, I shall start the procedure in two minutes.

Examiner: What did the pilot say?

Candidate: The pilot contacted the tower and informed it that he would start the procedure in two minutes.

(must -> must)

Center: Anac123, you must maintain FL090, report top of descent.

Examiner: What did the controller say?

Candidate: The controller told me that I must maintain FL090, and asked me to report top of descent.

Reported Questions

So now you have no problem with making reported speech from positive and negative sentences. But how about questions?

Direct speech: ATC: Anac123, Can you confirm that you suffered a tail strike?
How can we make the reported speech here?

In fact, it's not so different from reported statements. The tense changes are the same, and we keep the question word. The very important thing



though is that, once we tell the question to someone else, it isn't a question any more. So we need to change the grammar to a normal positive sentence. A bit confusing? Maybe this example will help:

Direct speech: ATC: Anac123, Can you confirm that you suffered a tail strike?

Examiner: What did the controller say?

Candidate: The controller asked me if I could confirm that I had suffered a tail strike.

Center: Anac123, have you requested FL090? Is this correct?

Examiner: What did the controller say?

Candidate: The controller asked me if I had requested FL090? and also asked if that was correct?

Adverbs

now -> then / at that time

today -> that day

yesterday -> the day before

tomorrow -> the day after

last night -> the night before

last week -> the week before / the previous week

tomorrow -> today / the next day / the following day

Vocabulary from previews exams

The following vocab is a must know because all of them came from preview exams and might be on yours. Do not make the mistake of not knowing it.

P1

at your discretion

air cabin depressurization

direct approach

expect approach

hot air balloon

holding point

give away to the aircraft



got stuck
disembark passengers
towed
reject take-off
engine stall
vacate rwy
need assistance
holding short
maintain position
rwy incursion
radar contact lost
under visual meteorological conditions
resume own navigation

P2

maintain position
hydraulic fluid leak
left main landing gear
via taxi L
taxi with caution
work in progress
expedite taxi
fuel leak
back track
line up and wait
return to the parking
unaccompanied minor
a person passed out
expect ILS approach
hot air babylon at 12 o'clock
proceed to AFA VOR
transmission is readable
transmissions is poor
readability is poor
distress call
thunderstorm
severe turbulence
dump some fuel



straight in approach
sudden decompression
altimeter setting
door separated from the airplane
airport authority
emergency evacuation
emergency services
scapes slides accidentally deployed
flight attendant
flight purser
stewardess

P3

reduce speed to 230 kt
exit hold now
squawk ident
transmission is blocked
cleared the arrival
expect hold
cleared straight in approach
continue approach
expect circling
downwind leg
rwy incursion
cross rwy
military activity
drift down procedure
maximum altitude capability
restricted area
struck by lightning
priority to land
vectors to final approach

P4

taxi to gate 5
report vacated
collision with a tow truck



maintenance people are on the way
fire fighters are on the way
start up at 60
standby frequency at 120.70
due to pressurization problem
reduce speed to mach 0.75
fly direct
confirm diversion
cleared to divert
confirm deviation
cleared to deviate
confirm detour due to bad weather
follow the traffic ahead of you
going around
main gear is stuck
proceed to AFA VORTAC
experiencing problems
confirm intentions
fire in the hold
emergency landing
descend at your discretion
assistance upon arrival
operations closed
in the vicinity of the airdrome
shower over the field
human organ for transplant
total electrical failure
battery power
number one for landing

P5

be advised
flock of birds on final
vultures in the vicinity
say indicated airspeed
copied you lost ailerons
line up and wait
the flight attendant has fainted



firefighter will be sent straight away
check out the problem
read back all information
confirm rwy in sight
on the visual
approaching holding point
follow acft ahead of you
hold position
ground staff on the way
looks like I have a failure
GPS inoperative
resume convention navigation
partition control of the ailerons
hydraulic pressure indication is dropping
perform an emergency landing
confirm if the approach is adequate

P6

received pilots reports
icing conditions
de-icing system not working properly
intend to divert
intend to perform ILS
rwy lights in sight
main landing gear collapsed
main landing gear retracted
main landing gear twisted
main landing gear extended
immediate takeoff
present heading
rate of descent
experience wind shear
we have a cracked windshield
we have a cracked windscreen
we have a shattered windscreen
maintain separation from proceeding acft
report base leg
it is opened in the front of the plane



it is opened in the aft cargo hold
it is opened in the forward cargo hold
leaving FL100
pilot incapacitated
pilot incapacitation
food poisoning
bird strike
smoke in the cabin
fumes in the cabin
hit birds
loss of thrust
loss of power
we lost engine number 1
leveled at flight level 200
disagreement in the altimeters

P7

attempt for subsequent takeoff
expect 40 minutes delay
hydraulic fluid leak
left main gear
expect vector to intercept
facing low fuel
facing fuel starvation
under radar radar vectors
broken windshield
approved taxi
loose dog on board
uncontrollable fire
steep descent
heavy vibration
engine surge
low oil pressure
dangerous goods on board
hazmat on board
ILS unserviceable
confirm people on board
confirm souls on board



P8

radio contact lost
loss of radio contact
passenger fighting on board
fire on board
passenger misbehavior
injured passengers on board
passengers misconduct
some passengers got hurt
cargo apron
taxi to the ramp
expect departure in about 3 minutes
follow missed approach procedure
maintain own separation
fly en route
message was blocked
landing gear stuck
landing gear bugged down
message was garbled
flaps up landing
belly landing
wheels up landing
flying northbound
due north
experiencing loss of power
neither of propellers is working
electronic navigation capability
we are assessing the system
Flight management system (FMS) went blank

P9

weather is deteriorating rapidly
heart attack
seizure
epileptic attack
microburst on final
windshear on final



gust wind on final approach
fire in the wheel well
monitor ground
debris on the taxiway
wreckage on the RWY
drone near the airport
FOD was reported
be prepared for right overtaken
TOD top of descent is close
confirm ETA (estimated time of arrival)
flame out on the right engine
an engine flame out
engine low oil pressure
engine high temperature
confirm endurance
confirm fuel remaining in hours

P10

cleared for immediate takeoff
maintain rwy heading
fire in the catering
start up and pushback approved
vectoring due traffic
smoke in the passenger cabin
overhead compartment
head up display HUD
request fuel dumping
restricted area
windscreen damaged
total electrical failure
partial panel
it is not feasible to maintain this altitude
at nearest suitable airport
at closest suitable airport

P11

expect RNAV RWY 20 approach



left hand pattern
pattern work
expect moderate turbulence
expect severe turbulence
shut engine down
high oil temperature
number of persons on board
push back tow truck
push back tow bar
deviation to avoid weather
longest available rwy

EVENTS, DOMAINS, AND SUBDOMAINS IN AERODROME CONTROL

Domains and words expected to be used

[Airmiss\(es\)](#)

Air traffic rules; avoiding action; trajectory/flight path; speed; distance/range; aircraft characteristics; position.

[Airshows Approach delays](#)

Traffic information; activity: acrobatics, formation flights; procedures.

[Belly landing](#)

Holding instructions; holding procedures; aerodrome circuit; endurance; diversion/alternate; necessary conditions; CAT III; all-weather landings. Attempted manoeuvres; status of lights; visual check (low pass); position of landing gear; endurance; fuel remaining; fuel dumping/jettisoning; speed; traffic information; state of runway; aerodrome environment; airport installations; emergency evacuation; emergency slides/escape chutes, etc.; fire hazard/risk; damage; ground services.

[Bird risk/hazard](#)

Position; quantity; names/types of birds; bird scaring in progress; damage to aircraft; delays; bird scaring methods; behaviour of birds.



Bomb threat/alert/scare

Disembarking passengers; diversion; baggage identification; fuel dumping/jettisoning; aircraft interior; crew actions/behaviour; ground services; airport installations.

Cargo problems/ dangerous goods

Customs; type of cargo; (perishable) organs for transplant; toxic substances; handling; packaging; veterinary services; police search; sniffer dogs; load badly fixed or damaged; intercepting; impounding.

Fire on board

Ground services; aircraft interior; smoke; asphyxia; smells; oxygen masks; warning lights; firefighting equipment; extinguishers; injuries; burns; medical assistance; fire brigade/firemen; emergency slides/escape chutes; engine shutdown; evacuation.

Ground movement incidents

Activity on the field; fire brigade training exercises and interventions; vehicles on the field; braking action and visibility; traffic information; start-up; towing equipment; engine checks; remote holding pattern; holding point; runway infringement; delays; stuck in the mud; damage caused by vehicles on the ground; no entry disregarded; collisions; vehicle or plane breakdown; damage to beacons; foreign objects (name, description); problems boarding or disembarking passengers; baggage identification; means of disembarking; health services; handicapped/sick passengers; parking position/space.

Health problems

Symptoms; first aid; aircraft interior; CPR, type of medical assistance; medical background of passengers; diversion; airport installations; ground services;



sickness; discomfort; wounds; epidemics; medical equipment; blood (group, transfusion, etc.); medical advice; the human body; forensic surgeon; quarantine; food poisoning; food; vaccines; medical staff; medicines and artificial limbs.

Incidents on landing

Long/short landing; missed exit; stuck in mud; weather; cargo problems; runway confusion; bird or animal hazard; damage to tires; aircraft breakdown; missed approach.

Industrial action

Ground staff; control/operational staff; effects on traffic; delays; types of strike; demonstrations; sit-ins.

MET (weather) conditions

ATIS (visibility, clouds, etc.); (thunder) storms, lightning; damage and breakdown; snow clearing; gusts; wind shear and microburst; minima; state of runway; tailwind, crosswind; braking action; runway visual range; temperature inversion; turbulence; natural disasters; runway closed; change of runway.

Missed approach

Go-around; minima; traffic position; endurance; reasons; traffic; procedures; speed.

Parachute jumping/ dropping activity

Position; information on other traffic and activity; duration of drop; drop zone.

Pilot not familiar with airfield

Procedures; airfield installations; ground services.

Pilot's temporary disability



Health problems; aircraft controls and instruments; pilot's actions/behaviour; airfield environment; airport installations.

Problems linked to flight plan

Delays; slots; flight plan updating; computer breakdown; no flight plan; flight plan conformity; flight plan processing; search and rescue; type of flight plan.

Problems linked to passenger's behaviour + unlawful interference

Violent/threatening behaviour; reasons (drunkenness, etc.); aircraft interior; damage; weapons; actions to overpower; police/fire rescue team assistance requested; demands; ethnic origin; physical description of person(s); political allegiances; ground services; airport installations; injuries/wounds; stowaways.

Re-routing/diversion Special flights

Approach charts; procedures; routing; endurance; weather; airport installations; ground services; aircraft breakdowns. ILS calibration; special test flight procedures; banners, balloons, etc.; ultra- lights, gliders; helicopters; aerial photography; highway watch; firefighting aircraft; supervision of power lines; military training procedures; types of military aircraft.

Take-off incidents

Abort; bird/animal hazards; traffic interference; runway incursion; overheating; towing; 180° turn back; runway excursion; cancellation and change of clearance; problems with steering gear, engine power; aircraft breakdown.

VFR flights lost/in difficulty

Aerodrome environment; direction finder; manoeuvres for identification; endurance problems; installations at alternate/diversion field; forced/crash landing; ground services.



VIP flights

Official ceremonies; protocol (greetings, etc.); ferry flight; military escort; diplomatic clearance; country names and nationalities; apron/ramp; terminal; boarding and disembarking of passengers; VIP vehicles; effects on traffic.

Administrative problems Aids for VFR flights

Diplomatic clearances; customs regulations; civil service departments; impounded aircraft.

Aircraft breakdowns

Instrument panel; on-board equipment; pilot rating; flight plan; local place name; visual landmarks; positions; directions; endurance; aircraft breakdown; weather problems. Instrument panel; instrument operation; radio beacon; positions/fixes; noises/sounds; smells; smoke; airport installations; ground services; engine performance; speed; relief/high ground; actions to solve problem; weather; fuel dumping/jettisoning; flight profile; structural damage (glass, metal); flight systems; aircraft controls; response to controls; airframe; warning lights; landing gear.

Aircraft proximity + pilot complaints

Conflict situations; traffic load; aircraft characteristics; flight profile; weather conditions; injuries; distance/range; pilot manoeuvres; rules; procedures; avoiding action.

ATC system breakdowns

ATC equipment/systems; radar display; radar performance; radio operation; previous messages; relaying messages; actions to repair; delays/duration; telephone lines.

Bomb scare



Aircraft interior; search methods; fuel dumping/jettisoning; ground services; airport installations; ground movements.

[Cargo problems/ dangerous goods](#)

Packaging; substances; toxic substances; animals; smells; cabin equipment; load distribution; loading/unloading.

Change in flight plan or Collisions

Airframe; structural damage (glass, metal, etc.); response to controls; debris; airport installations; ground services; relief/high ground; weather conditions; aerodynamic behaviour.

Fire on board Health problems

Outbreak of fire; control of fire; damage; aircraft interior. Parts of the body; organs; symptoms; sicknesses; injuries/wounds; artificial limbs; medicine/drugs; first aid; medical equipment; medical staff; medical specialists; vaccines; quarantine.

Lack of fuel

Airport facilities/installations; ground services; high ground; positions/locations; endurance/fuel remaining, running low on fuel.

Misunderstandings

Previous messages; types of messages; radio performance, clarification, ready back, hear back, say again.

Passenger behaviour + unlawful interference

Violent threatening behaviour; drugs; firearms; injuries; mental instability; nationalities; political allegiances; demands, threats; ground services; medical assistance; means of calming; means of overpowering; flight deck and cabin personnel.

Request to relay



Means of relaying, relaying requests, relaying a message.

Special conditions on arrival

State of the traffic on the ground; priority flights; industrial action; accidents; weather conditions on the ground; ground equipment failure; airport installations; ground services; curfew; approach procedures.

Special flights

State of the traffic on the ground; priority flights; industrial action; accidents; weather conditions on the ground; ground equipment failure; airport installations; ground services; curfew; approach procedures.

Unauthorized manoeuvres Weather/MET problems

Type of aircraft; ferrying; diplomatic personnel; country names; nationalities; aeronautical military slang; military exercises; in-flight/mid-air refuelling; pilot manoeuvres; positions/fixes; weather conditions; VFR/IFR procedures; visual flight rules; airport installations; ground services.

Activities on the field

Airspace; rules; previous messages; flight profile; positions/locations; stall levels.

Aerodrome/airfield environment

Icing problems; clouds; struck by lightning; turbulence; external parts of aircraft; engine performance; response to controls; instrument performance; alarms; violent movements; relief/high ground; flight profile; injuries; blindness/loss of visibility, Airfield facilities/ installations.



3. OTHER DOMAINS

Change of runway and pattern; ramp vehicles; snow clearing; sweeping; mowing; harvesting; closure; opening of runway access roads; runway inspection, snowplough, light crane, heavy crane.

Topography (hill, slope, coastline, forest, etc.); civil engineering (water, tower, bridge, pylon, etc.); high ground/terrain; built-up areas; roads and railway lines; power lines; cardinal points; particular local activities (firing range, etc.); agricultural activities.

Aircraft spare parts; systems (oxygen, hydraulic, electrical, de-icing, etc.); flight deck/cockpit; controls; instruments; instrument operation; noises and symptoms of malfunction; transponder problems; loss of radio contact; malfunctions; overheating (brakes, engine, etc.); fuel dumping/jettisoning; landing gear/tires, tire blow out.

ILS, radar, VOR, etc.; lighting systems; reliability of radio aids; direction finder; poor visibility equipment; aprons/tarmac/ramps; runways; taxiways; length and width of runway; parking zone; holding area; terminal; cargo area; bearing strength.

Opening hours; availability of services at night; assistance on the ground; safety altitude; passengers/persons on board; unserviceable equipment (stairs, luggage trolleys, etc.); auxiliary power unit; de-icing; refuelling; delay due to de-icing or refuelling; bird scaring; towing; firefighting methods; safety services; medical assistance; baggage handling.

Noise abatement; departure; approach; all-weather take-off and landing; go-around; holding procedures; land use; curfew; local residents.



Check some reported speech examples:

Situation 1: ANAC 123, stop your descent at five thousand feet and reduce to minimum clean speed. Are you requesting to land on runway one two left? Confirm.

What did the controller say?

First, the controller asked me to stop my descent at five thousand feet and also to reduce to minimum clean speed. And finally he asked me to confirm if I was requesting to land on runways one two left.

Situation 2: ANAC 123, descend at your discretion to flight level zero seven zero, turn right heading zero seven zero. No reported traffic. Confirm you have an air cabin depressurization.

What did the controller say?

The controller cleared me to descend at my discretion to flight level zero seven zero and turn right heading zero seven zero. He also informed me that no traffic was reported and finally asked me to confirm If I had an air cabin depressurization.

Situation 3: ANAC 123, fly direct to PIR VOR and expect direct approach to runway zero six left. I understand you lost your right engine. Confirm?

What did the controller say?

The controller cleared me to fly direct to PIR VOR and to expect direct approach to runway zero six left. And said that he understood I had lost my right engine. And, in the end, asked me to confirm?

Situation 4: ANAC 123, turn three zero degrees left. Confirm the hot air balloon is at your three o'clock position.

What did the controller say?



The controller instructed me to turn three zero degrees left, and asked me to confirm if the hot air balloon was at my three o'clock position.

Situation 5: ANAC 123, I understand you left the taxiway and got stuck. Confirm the situation and if you'll need to disembark the passengers before being towed back onto the taxiway.

What did the controller say?

The controller told me he understood I left the taxiway and had got stuck. After he asked me to confirm the situation and if I would need to disembark the passengers before being towed back onto the taxiway.